
ALGOJAM 3 · ROUND 1

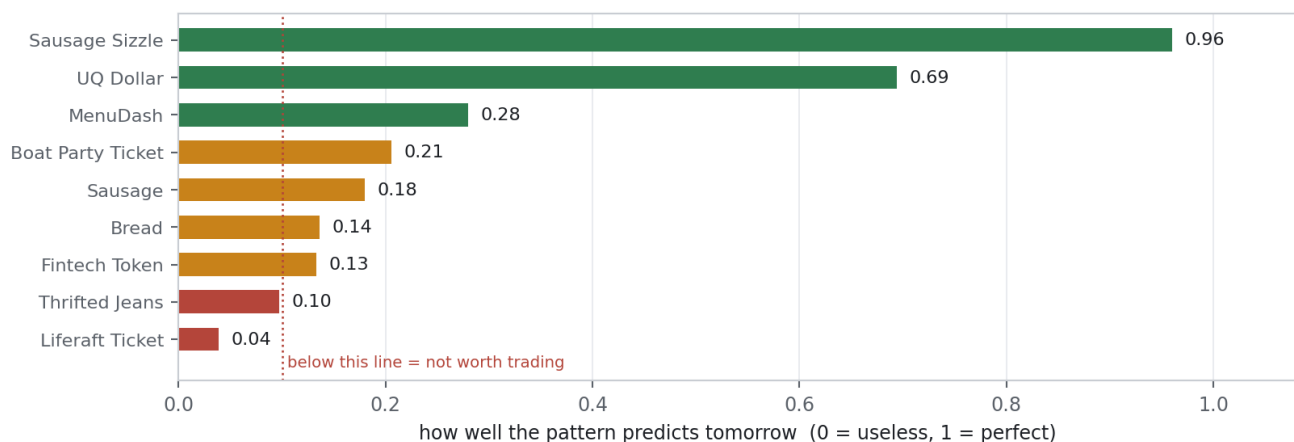
What each instrument actually is

A plain-English guide to the nine price series: which ones hide a rule you can trade, which ones are coin flips wearing a costume, and how you tell the difference.

The one-page answer

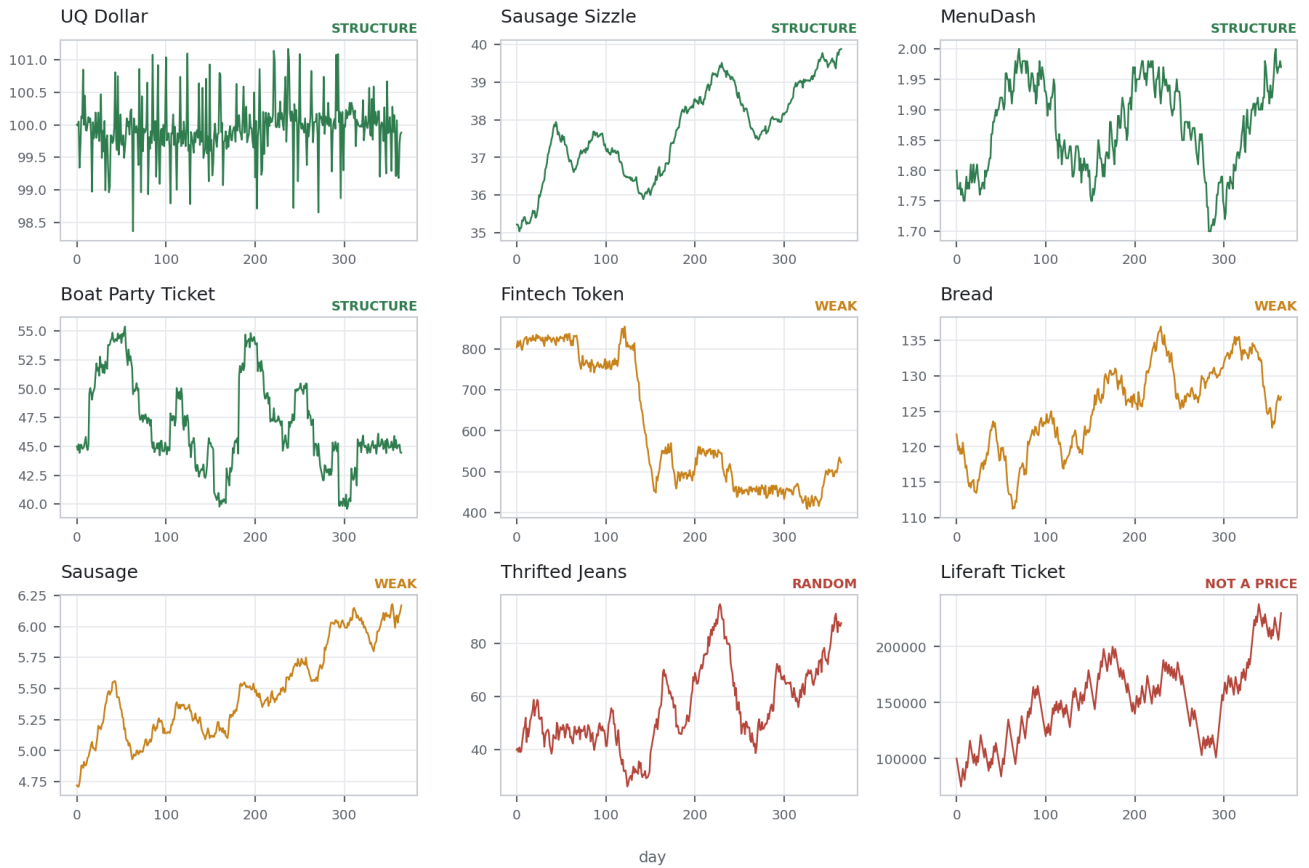
Four instruments have a real, repeatable rule. Three have a faint edge. Two have nothing you can trade.

INSTRUMENT	WHAT IT REALLY IS	VERDICT
Sausage Sizzle	A recipe. Its price is literally bread + sausage + labour, using <i>yesterday's</i> ingredient prices.	SOLVED
UQ Dollar	Pegged to \$99.94 by an invisible elastic band. Snaps back within a day, every time.	SOLVED
MenuDash	A blurry, 1-cent-rounded read of the same labour cost. Drifts away, gets pulled back.	STRUCTURE
Boat Party Ticket	A crowd. It panics, overshoots, and bounces back over about three days.	STRUCTURE
Sausage	Random walk with a mild upward drift and slight trend-following.	FAINT
Bread	Random walk. Slight tendency to keep going in the same direction.	FAINT
Fintech Token	Random walk in direction, but its <i>volatility</i> is highly predictable.	FAINT
Thrifty Jeans	A pure random walk. Big year, no rule. Statistically a coin flip.	RANDOM
Liferaft Ticket	Not a price at all — it's the scoreboard of a vote between teams.	NO SIGNAL



How strongly each pattern predicts tomorrow's move. Measured causally (only past data), then checked on the first and second halves of the year separately. Anything under 0.10 could be luck.

The nine, at a glance

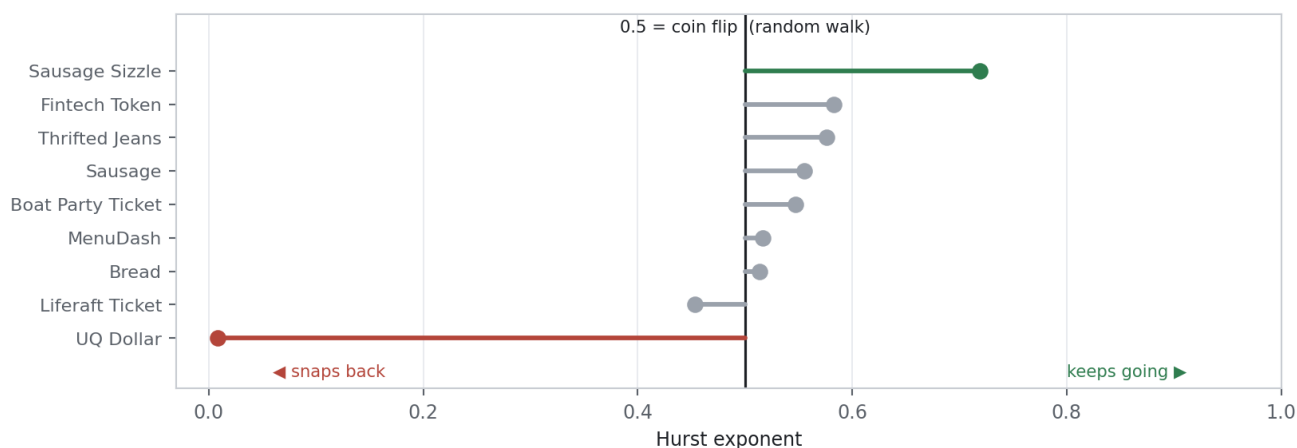


How to tell a random walk from a rule

A random walk is a price that takes an independent coin-flip step every day. Today's move tells you nothing about tomorrow's. Two simple measurements separate them from everything else.

Test 1 — does the price snap back, drift on, or neither?

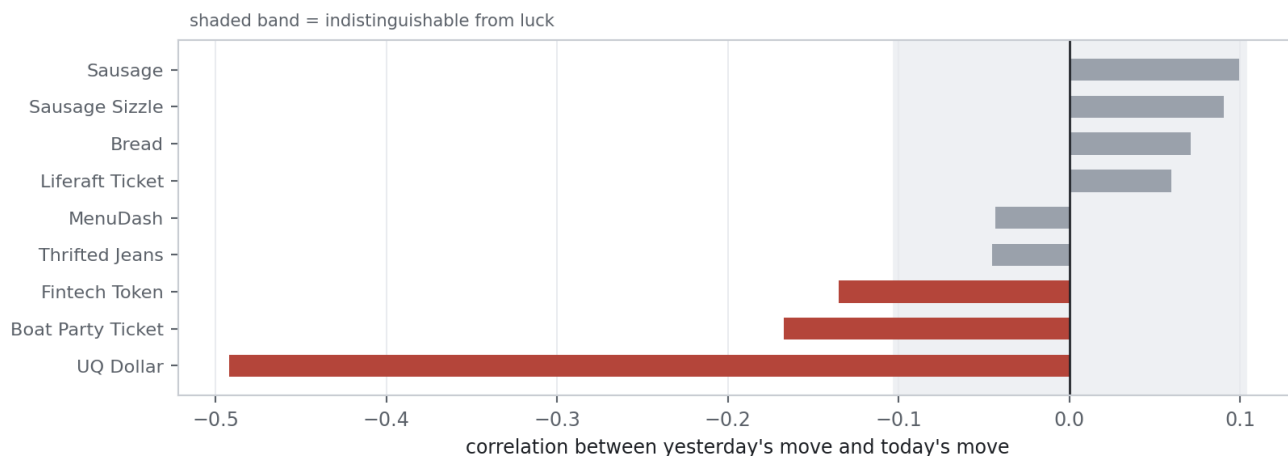
The Hurst exponent is one number on a 0–1 scale. **0.5 means coin flip**. Below 0.5, moves get reversed. Above 0.5, moves get extended.



Only two instruments are meaningfully off 0.5. UQ Dollar sits at almost zero — it reverses essentially every move. Sausage Sizzle sits at 0.72, but that is the slow ingredient trend leaking through, not a trend you can trade directly.

Test 2 — does yesterday's move predict today's?

For a coin flip, this correlation is zero. With 365 days, anything inside ± 0.10 is indistinguishable from luck.



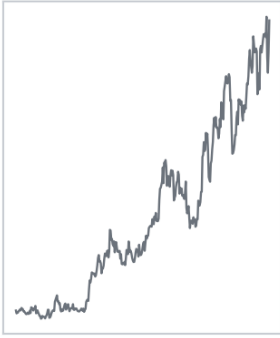
UQ Dollar at -0.49 is the standout: half of yesterday's move is undone today. Boat Party and Fintech Token show mild overreaction. Bread, Sausage and Sizzle lean the other way — mild trend. Thrifted Jeans and Liferaft sit inside the noise band.

The sanity check

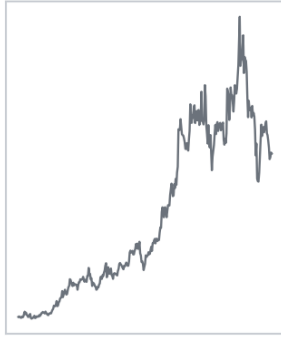
Below are four price paths. Three were generated by a computer shuffling coin flips. One is the real Thrifted Jeans series.

Three are computer-generated coin flips. One is Thrifted Jeans.

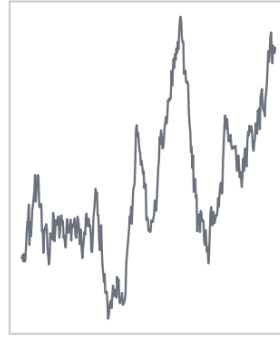
A



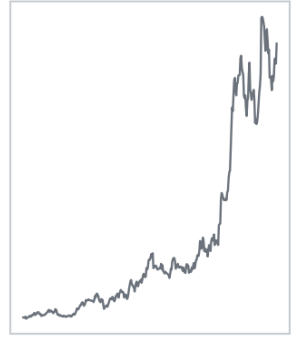
B



C



D



C is real. If you could not tell, that is the point — Thrifted Jeans is statistically indistinguishable from a coin flip. Its 119% gain over Round 1 was one draw of the dice, not a pattern.

Sausage Sizzle

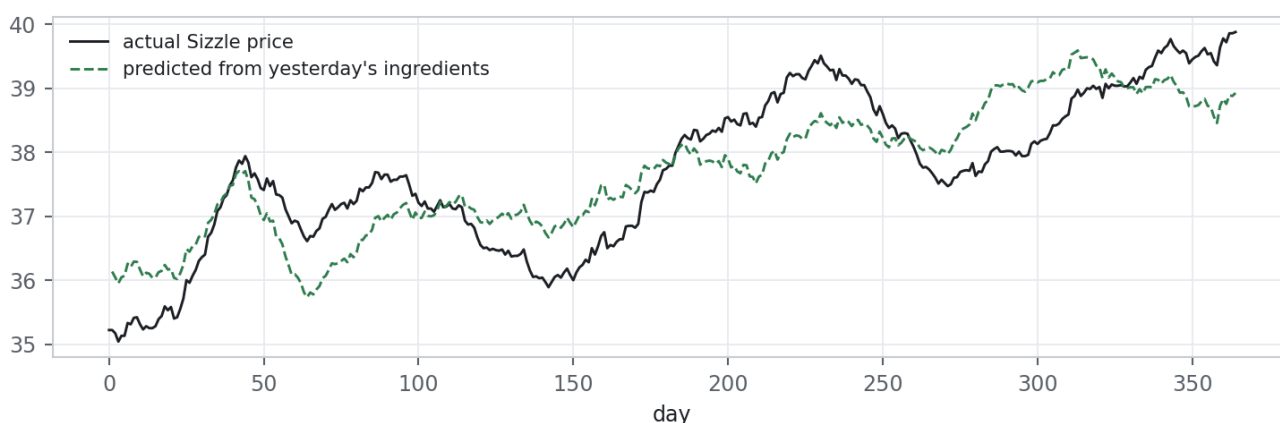
SOLVED — STRONGEST EDGE ON THE BOARD

You are not forecasting a price. You are doing arithmetic on numbers that are already printed.

The sizzle's price is its cost of goods. Fit it and you get an almost exact recipe, and critically it uses *yesterday's* ingredient prices — whoever runs the stall shops the day before.



Bread and Sausage print first — so tomorrow's Sizzle price is already 96% known before the day starts.



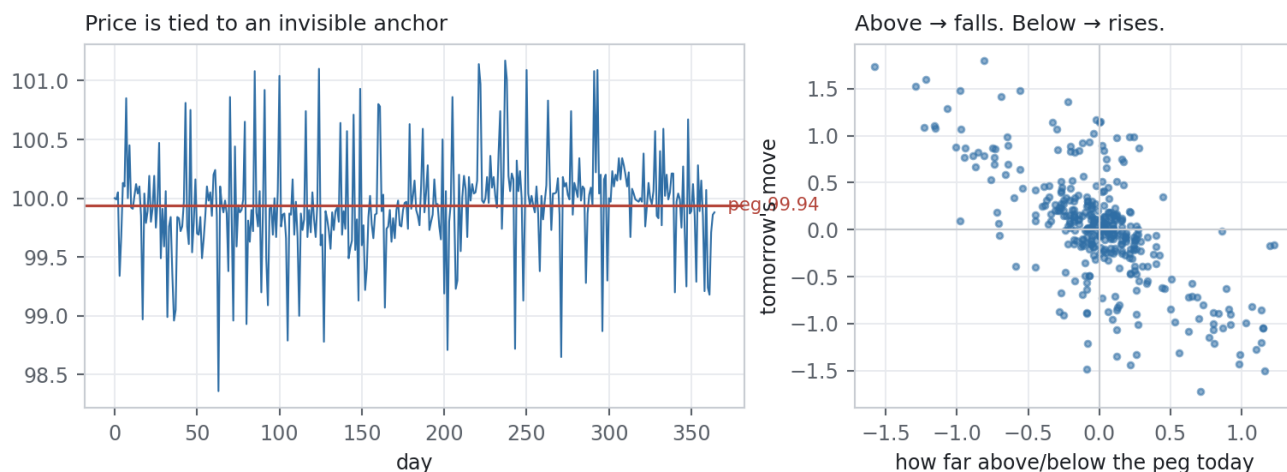
The dashed line is the price predicted from yesterday's bread and sausage alone, using one fixed set of coefficients for the whole year. It follows every turn. The small level gaps are the slowly-moving labour term, which a rolling fit removes. $R^2 = 0.70$ on levels; on day-to-day changes the prediction lines up with reality at 0.96.

What this means in practice

Every morning, bread and sausage prices for yesterday are known. Multiply, add the labour term, and you know the direction of tomorrow's sizzle move with near-certainty. Take the full position on that sign. Don't waste it on a momentum signal.

It is tied to \$99.94 with a very short elastic band.

The price wanders off the peg during a day and is dragged back almost entirely by the next. The half-life of a deviation is **0.72 days** — more than half of any gap closes within 24 hours.



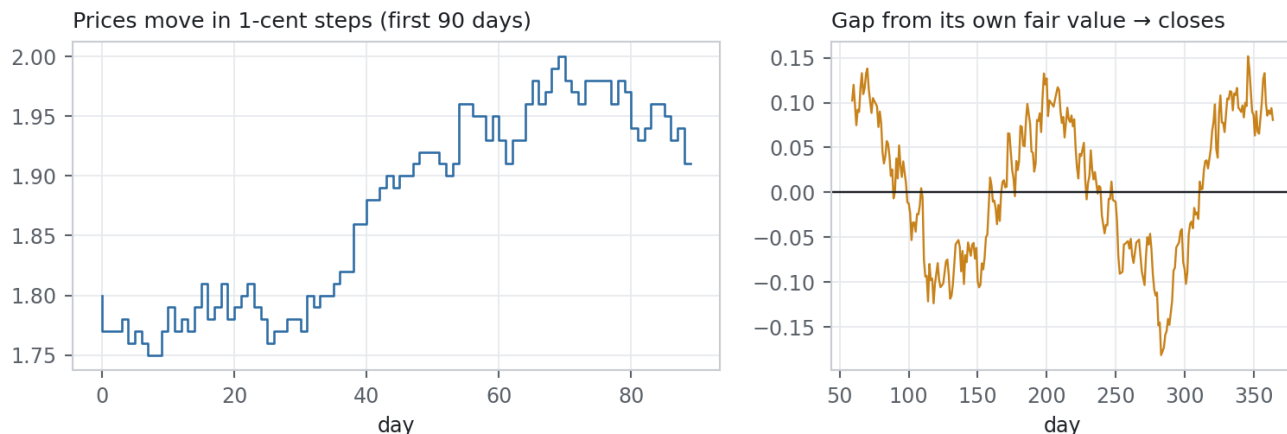
Left: every excursion returns. Right: each dot is one day. The downward slope is the whole trade — the further above the peg it sits today, the harder it falls tomorrow.

What this means in practice

Buy below 99.94, sell above it, size in proportion to how far away it is. Small gaps still work: deviations of 0.05–0.15 are right about 60–70% of the time, and gaps over 0.33 are right 92–95% of the time. A hard on/off deadband throws that gradient away.

A rounded, noisy read of the same labour cost that sits inside the Sausage Sizzle recipe.

Its level tracks implied labour cost with a correlation of 0.93. When it drifts away from that fair value, the gap closes in about three days. Two quirks matter: prices move only in whole cents, and on 22% of days they don't move at all.



Left: the 1-cent grid, visible when you zoom in. Right: the gap between price and its own rolling fair value — it oscillates around zero rather than wandering off.

What this means in practice

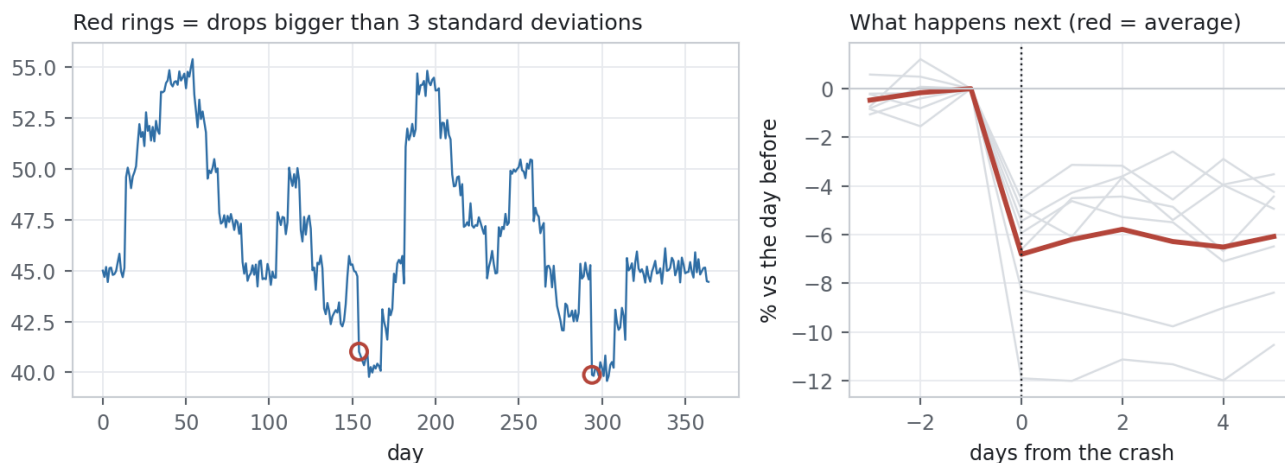
Trade the gap to fair value, not the raw price. But require the gap to exceed about one cent before acting — sub-tick gaps are noise, and this instrument already has by far the worst turnover on the board.

Boat Party Ticket

STRUCTURE — OVERREACTION

The crowd instrument. It panics harder than it should, then takes it back.

This is the only series with genuinely fat tails: three moves larger than four standard deviations, seven above three. After a big drop the next day averages **+0.24**; after a big spike, **+0.12**. The bounce is real and it is stronger after crashes.



Right: every large drop in the year, lined up on the day it happened (day 0) and indexed to 100 the day before. Grey lines are individual episodes, red is the average. The average falls about 7% on the day, then grinds back up over the following days. The bounce is small per episode but it shows up nearly every time.

What this means in practice

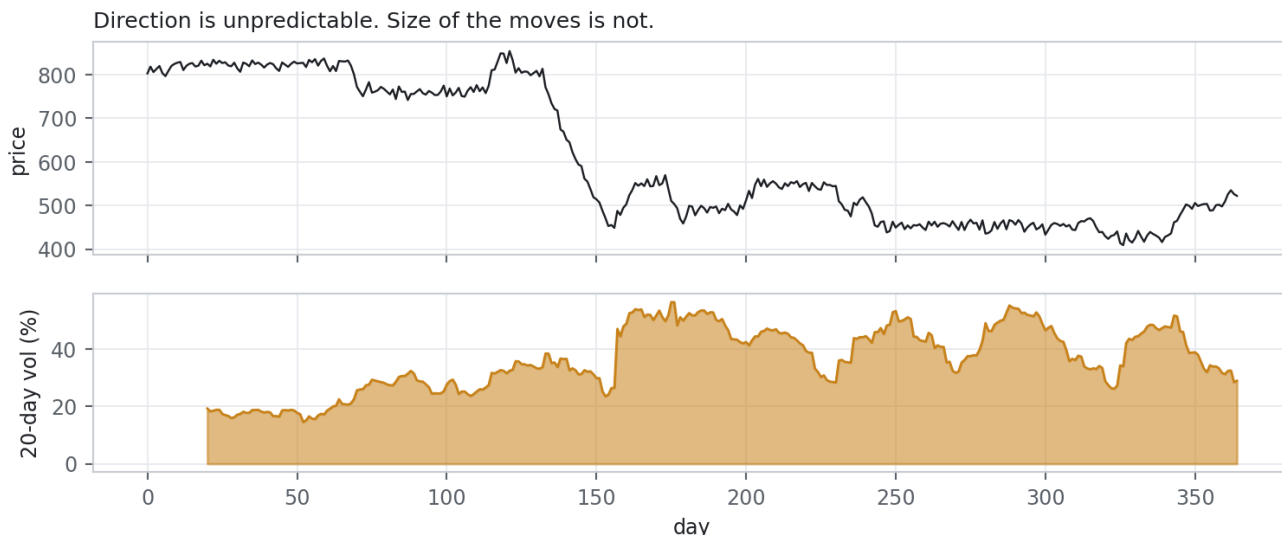
Fade three-day moves. Buy weakness, sell strength.

One warning

The price looks seasonal — two semester-shaped humps. That shape does *not* transfer. Fitting the calendar pattern on the first half of the year and applying it to the second half gives a signal that points the *wrong way*. Any hard-coded day-of-year table is fitted to Round 1's particular calendar and is a liability in Round 2. Use a rolling trailing slope if you want trend, or drop the seasonal term.

Where the price goes is a coin flip. *How far* it goes is not.

This is the only instrument with real volatility clustering: quiet stretches follow quiet stretches, wild stretches follow wild ones. Rolling 20-day volatility swings by a factor of 2.7 across the year and is 97% correlated with itself day to day — it takes about 47 days to fade back to normal.



Top: the price, which is a random walk. Bottom: how volatile it has been. The bottom panel is forecastable; the top panel is not.

What this means in practice

Put the volatility forecast into your *position size*, not your direction. Trade smaller when volatility is high. For direction, only a 1-2 day reversion holds up; the 5, 7 and 10-day versions point one way in the first half of the year and the opposite way in the second, which means they are fitted noise.

Bread and Sausage

FAINT — MILD TREND

Both are random walks with a gentle tendency to keep going.

Sausage drifts upward over the year and mildly follows its own trend; bread does the same, weaker. Both hold up when you split the year in half, so the edge is probably real — it is just small.

INSTRUMENT	SIGNAL	STRENGTH	1ST HALF	2ND HALF
Sausage	11-day momentum	0.18	0.26	0.07
Bread	10-day momentum	0.14	0.13	0.14

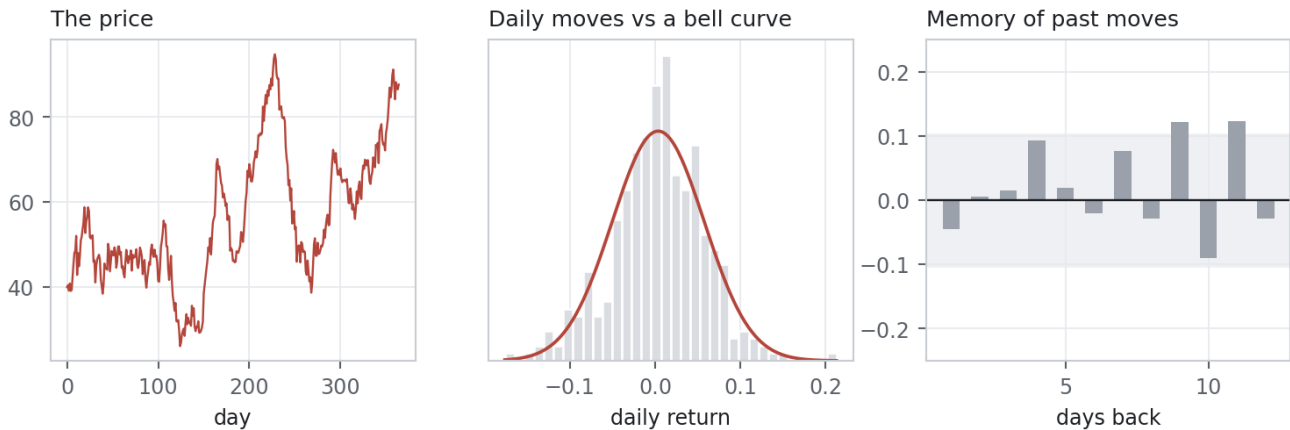
Bread's edge is the more trustworthy of the two — identical in both halves. Sausage's weakens noticeably in the second half.

What this means in practice

Keep the momentum signals but fund them last. They earn the least per dollar of budget on the whole board, and Sausage in particular churns a lot of volume for it. Their real value is upstream: they are the inputs to the Sausage Sizzle forecast.

The clearest coin flip in the set. It gained 119% in Round 1 and none of that was predictable.

Every test comes back empty. Hurst 0.58, yesterday-to-today correlation -0.05 (inside the noise band), no volatility clustering, normal-shaped returns. The momentum signal that looks tempting scores 0.097 with a t-statistic of 1.82 — below the bar, and exactly what you would expect to find by chance after testing a few window lengths.



Left: a dramatic-looking year. Middle: the daily moves sit under a textbook bell curve — no fat tails, no surprises. Right: the correlation with its own past, out to twelve days. Every bar is inside the grey luck band. There is nothing there.

What it does have is **86% annualised volatility**, the highest of any real price series here. A one-year gain of 119% on that kind of volatility is well within the range of luck. The same generator could just as easily print -50% in Round 2.

What this means in practice

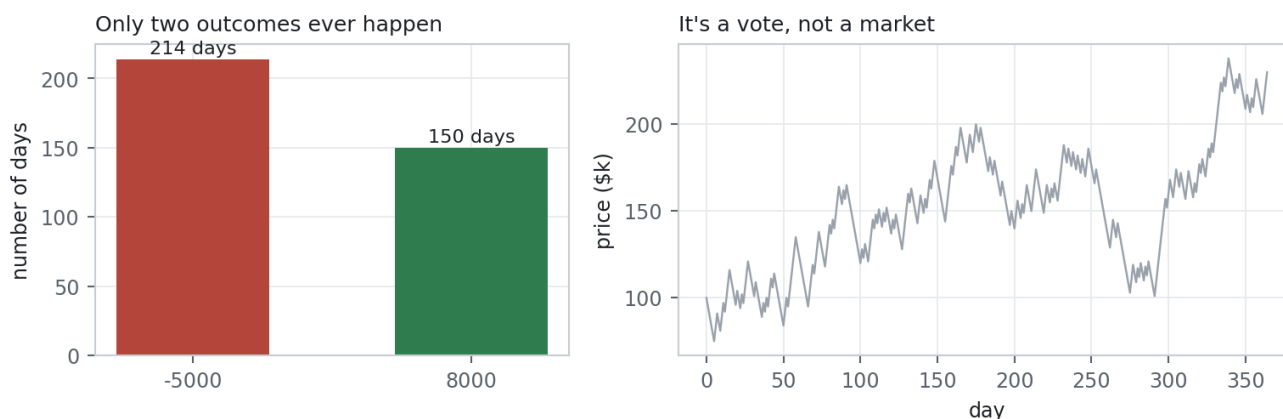
There is no signal to trade. A partial long is a defensible bet on the drift continuing, but be clear that it is a bet, not a model. Size it accordingly.

A note on the other random walks

Bread, Sausage and Fintech Token are also random walks in the strict sense — their prices pass the standard tests for one. The difference is that each has a small secondary feature layered on top (mild trend, mild trend, volatility clustering) that survived split-half testing. Thrifty Jeans has nothing layered on top at all.

This one isn't a market. It's a vote, and Round 1's history tells you nothing about Round 2's room.

The daily price change takes exactly two values and never anything else: **−5,000** on 214 days and **+8,000** on 150 days. That is a payoff table, not a price series. The outcome each day is decided by which way the majority of teams positioned themselves.



Left: the only two moves that ever occur. Right: the resulting path — which is just a record of how one particular room of competitors behaved.

Why the history is useless

Round 1's sequence reflects Round 1's competitors. Consecutive days are independent (correlation 0.04, average run length 2.1 — exactly what independent flips give). A contrarian rule fitted to it scores -0.04 , i.e. worse than nothing.

The only defensible play

Use the payoff asymmetry, not the history. Going long wins +8,000; going short wins +5,000. So long is the better side whenever fewer than **61.5%** of teams go long. Round 1's rate was 58.8% — just under the line. Stay long, and treat that as a prior about human behaviour, not a signal from the data.

Things that look like structure and aren't

Four patterns that survive a first glance and die under testing. Each one would have cost money.

1. The Boat Party seasonal calendar

83% of its variance is a smooth two-humped shape that looks exactly like two university semesters. Fit the shape on days 0–181 and apply it to days 182–364 and it predicts backwards. Round 1's two semesters simply aren't copies of each other.

2. Cross-sectional mean reversion

Rank the nine each day, buy the losers, sell the winners — the classic factor trade. Across all nine it tests at $t = -5.3$, which looks compelling. But that is entirely UQ Dollar's own violent reversion leaking into the pool. Remove the instruments that already have their own documented reversion and the effect disappears ($t = -0.98$). **There is no cross-sectional factor here.** Trade each instrument on its own model.

3. Volatility clustering in Bread, Sizzle and MenuDash

A GARCH model reports persistence of 0.98 on all three, which would normally mean strong clustering. The formal test for it says $p > 0.30$ — there is no clustering. The model is mistaking the slow ingredient-cost trend for persistence. Only Fintech Token and UQ Dollar have genuine volatility structure.

4. Cointegrated pairs

UQ Dollar appears to pair with all eight others at $p < 0.001$. That is an artefact of UQ Dollar being stationary on its own, not evidence of a relationship. Of the 36 pairs tested, only the two Sausage Sizzle ingredient links survive a correction for multiple testing.

The general lesson

With 365 observations, a t-statistic of 2.5 is weak evidence. The test that actually separated the real patterns from the fitted ones was the simplest: fit on the first half of the year, check on the second, and throw away anything that changes sign.

What to trust going into Round 2

Round 2 is a fresh draw from the same generators. The *structure* should carry over — the peg, the recipe, the cent-rounding, the overreaction. The *fitted numbers* may not: drift rates, the seasonal table, and the profit-per-instrument rankings are all properties of Round 1's particular path.

All figures generated from the Round 1 price files. Full statistical detail, test methodology and rejected-hypothesis log: AlgoJam3_Structural_Analysis.pdf.